

```

1 import components.simplereader.SimpleReader;
2
3
4
5
6 /**
7  * To calculate the square root.
8  *
9  * @author Eric Zhou.
10 *
11 */
12 public final class Newton2 {
13
14     /**
15      * No argument constructor--private to prevent instantiation.
16      */
17     private Newton2() {
18     }
19
20     /**
21      * Computes estimate of square root of x to within relative error 0.01%.
22      *
23      * @param x
24      *         positive number and zero to compute square root of
25      * @return estimate of square root
26      */
27     private static double sqrt(double x) {
28         final double ep = 0.0001; //set the error
29         if (x == 0) { //solve the problem that x equals to zero
30             return x;
31         } else {
32             double r = x;
33             while ((Math.abs(r * r - x) / x) > ep * ep) { //calculate the square root
34                 r = (r + (x / r)) / 2;
35             }
36             return r;
37         }
38     }
39
40     /**
41      * Main method.
42      *
43      * @param args
44      *         the command line arguments
45      */
46     public static void main(String[] args) {
47         SimpleReader in = new SimpleReader1L();
48         SimpleWriter out = new SimpleWriter1L();
49         do {
50             out.println("Enter a non-negative number: "); //ask for the number
51             out.println(sqrt(in.nextDouble()));
52             out.println("Do u want to continue? Press 'y' or anything else to stop");
53         } while (in.nextLine().equals("y")); //check whether the user want to continue
54         /*
55          * Close input and output streams
56          */
57         in.close();
58         out.close();
59     }
60 }
61 }
62

```